

Nikita Kazeev

kazeevn@gmail.com · [Website](#) · [GitHub](#) · [LinkedIn](#) · [Google Scholar](#) · [ORCID](#)

Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

PERFOMAX *AI Advisor* 2026–present

NATIONAL UNIVERSITY OF SINGAPORE *Postdoc under [Kostya Novoselov](#)* 2022–present

Leadership

- Program Chair for the [AI4X 2025/6 conferences](#).
- Main organizer of the [ICLR 2025 Workshop on Machine Learning Multiscale Processes](#).

Wyckoff Transformer for generating symmetric crystals [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

CERN [YANDEX → HSE UNIVERSITY] *Researcher* 2014–2022

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

EDUCATION

Higher School of Economics (HSE University) <i>PhD in Computer Science</i> , supervisor Andrey Ustyuzhanin ; Machine Learning for particle identification in the LHCb detector .	2016–2020
Sapienza – Università di Roma <i>PhD in Physics</i> (double degree with HSE), supervisor Barbara Sciascia .	2016–2020
Product Management course at Yandex Focused coursework in product strategy and execution.	2018
Yandex School of Data Analysis Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.	2013–2015
Moscow Institute of Physics and Technology <i>MS in Physics</i> (2014–2016): Optimisation of data processing of the LHCb experiment. <i>BS in Physics</i> (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.	
International Junior Science Olympiad (IJSO) Silver medal (held in Korea).	2008

MENTORSHIP

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

TEACHING & OUTREACH

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
- Delivered more than 20 [conference talks](#).

SERVICE

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

SKILLS

- **ML & AI** – Structured, non-text, domain-specific data
- **Python** – Research coding
- **PyTorch** – Deep learning
- **Research Writing & Speaking**
- **Leadership & Mentorship**

SELECTED PUBLICATIONS

1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). Forecasting as a New Frontier of Intelligence Workshop at ICML 2026
2. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. [Position: Align AI to Our Aspirations, Not Our Flaws](#). Pluralistic Alignment Workshop at ICML 2026
3. Kazeev, Nikita and Ustyuzhanin, Andrey. [The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories](#). Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026
4. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
5. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
6. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
7. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. Alphabetic order, I'm the corresponding author.